Alkane

141. The marsh gas detector used by miners works on the principle of

(a) Difference in the rates of diffusion of gases

(b) Avogadro's hypothesis

(c) Gay-Lussac's law of gaseous volumes

(d) Berzelius hypothesis

142. Methane can be prepared by

(a) Wurtz's reaction

(b) Decarboxylation

(c) Hydrogenation reaction

(d) All of these

143. The most strained cycloalkane is

(a) Cyclopropane

(b) Cyclobutane

(c) Cyclopentane

(d) Cyclohexane

144. Which does not react with chlorine in dark

(a) C_2H_4 (b) C_2H_2

(c) CH_4 (d) CH_3CHO

145. Main constituent of *marsh gas* is

(a) C_2H_2 (b) CH_4

(c) H_2S (d) *CO*

146. Which of the following method can be used for the preparation of methane

(a) Wurtz reaction

(b) Kolbe's reaction

(c) Reduction of alkyl halide

(d) Hydrogenation of alkene

147. Which hydrocarbon will be most stable

(a) Methane (b) Ethane

(c) Propane (d) Butane

148. *C-H* bond length is greatest in

(a) C_2H_2 (b) C_2H_4

(c) C_2H_6 (d) $C_2H_2Br_2$

149. Which one of the following compounds does not form an ozonide

(a) Ethene (b) Propyne

(c) Propene (d) Propane

150. Which type of hybridisation occurs in ethylene

(a) *sp* (b) sp^2

(c) sp^3 (d) sp^3d

151. Silver acetylide when heated with HCl gives

(a) C_2H_2
(b) H_2
(c) C_2H_4
(d) None of these

152. When sodium reacts with ethyl iodide, which of the following hydrocarbons is produced
(a) Methane (b) Ethane
(c) Butane (d) Ethene

153. Solid methane is
(a) Molecular solid
(b) Ionic solid
(c) Covalent solid
(d) Not possible

154. The shape of ethane is
(a) Triangular
(b) Tetrahedral
(c) Linear
(d) None of these

155. CH_3MgI will give methane with
(a) C_2H_5OH
(b) $CH_3-CH_2-NH_2$
(c) $CH_3-CO-CH_3$ (d) All of these

156. Propane-1-ol can be prepared from propene by its reaction with
(a) CH_3COOH
(b) H_3BO_3
(c) $B_2H_6/NaOH, H_2O_2$
(d) H_2SO_4/H_2O

157. The process in which higher hydrocarbons are broken down into lower hydrocarbons by controlled pyrolysis, is called
(a) Hydrolysis (b) Cracking
(c) Oxidation (d) Reduction

158. Successive alkanes differ by
(a) $>CH_2$ (b) $>CH$
(c) $-CH_3$ (d) C_2H_4

159. General formula of alkane is
(a) C_nH_{2n+2} (b) C_nH_{2n-1}
(c) C_nH_{2n} (d) C_nH_{2n+1}

160. Methane and ethane both can be prepared in one step by which of the following compound
(a) C_2H_4
(b) CH_3O
(c) CH_3Br
(d) CH_3CH_2OH